S181M116 Derivatives and Alternative Investments

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Course Format

Main text	John C. Hull, *Options, Futures, and Other Derivatives*, 11th Global Edition (no need to buy the book, I will provide slides and lecture notes)
Course materials	GitHub course materials repository
Assessment	Assignment: 20%; closed-book final examination: 80%
Before attending	I will release lecture notes for the specific class one day before. Students should read them before attending otherwise they will not follow the material. This is a high paced, mathematically intensive course

Course Description

This intensive course introduces the economics, mathematics, and practical use of derivative securities and selected alternative investments. The core of the course follows Hull's treatment of forwards, futures, swaps, options, no-arbitrage valuation, binomial trees, Wiener processes, Itô's lemma, the Black–Scholes–Merton model, option Greeks, volatility, and risk management. Because the course also covers alternative investments and private equity, the final part of the course connects option-pricing logic to real options, venture capital, private equity fund economics, illiquidity, leverage, and performance measurement.

The course is designed for students who need both basic working knowledge and mathematical detail. The emphasis is not only on knowing the vocabulary of derivative markets, but also on being able to derive prices, state no-arbitrage arguments, interpret hedging equations, and explain why the mathematics matters for financial decisions.

Prerequisites

Students should be comfortable with algebra, time value of money, basic probability, normal and lognormal distributions, and introductory calculus. Prior exposure to finance is helpful but not essential. No prior stochastic calculus is assumed; the course introduces the necessary stochastic process tools directly.

Learning Objectives

By the end of the course, students should be able to:

1. Explain the economic role of forwards, futures, swaps, options, and selected alternative investments.
2. Distinguish exchange-traded and OTC derivatives, including margin, clearing, counterparty risk, and settlement.
3. Price forward and futures contracts using no-arbitrage cost-of-carry arguments.
4. Construct hedges using futures and explain basis risk, cross hedging, hedge ratios, and rolling strategies.
5. Use zero rates, forward rates, discount factors, duration, and convexity in derivative valuation.
6. Derive option bounds, put-call parity, and basic static option strategies.
7. Price European and American-style options with one-step and multi-step binomial trees.
8. Define Brownian motion and geometric Brownian motion and apply Itô's lemma.
9. Derive and interpret the Black–Scholes–Merton partial differential equation.
10. Apply risk-neutral valuation to obtain Black–Scholes–Merton option prices.
11. Compute and interpret Delta, Gamma, Theta, Vega, and Rho.
12. Explain implied volatility, volatility smiles, and why market prices deviate from the simplest Black–Scholes assumptions.
13. Connect derivative pricing concepts to real options, private equity, venture capital, leverage, and alternative investment risk.

Assessment

Assignment: 40%

The assignment will test both numerical competence and economic interpretation. It will be released after the first week and submitted before the final examination unless otherwise announced. All submitted work must explain the financial reasoning and mathematical steps.

Final Examination: 60%

The final examination is closed book. Students should expect a mixture of conceptual questions, derivations, and numerical problems. The exam may include formula recall, but the emphasis will be on understanding and applying the formulas rather than mechanically reproducing them. Unless explicitly stated by the instructor, books, notes, slides, solved exercises, and electronic communication devices are not permitted.

The final examination will cover all lecture material, all assigned Hull readings, and the private equity / alternative investment material distributed in class.

Main Reading Map from Hull

Core	Chapters 1–7, 10–15, 17–21, 35–37
Highly relevant	Chapters 19–23 for Greeks, volatility, numerical methods, VaR, and volatility estimation
Selective use	Chapters 8–9 and 24–26 for securitization, XVA, credit risk, credit derivatives, and exotic options
Alternatives bridge	Chapter 36 on real options; instructor-supplied notes for private equity, hedge funds, infrastructure, and illiquid assets

Detailed Two-Week Schedule

Week 1: Foundations, Forwards, Futures, Interest Rates, and Option Basics

Class 1: Derivative Markets, Payoffs, and No-Arbitrage Thinking

Hull readings: Chapters 1 and 2; skim Chapter 37 for examples of misuse and risk.

Topics:

- What derivatives are: forwards, futures, swaps, calls, puts, and structured exposures.
- Exchange-traded versus OTC markets.
- Long and short positions; payoff diagrams and profit diagrams.
- Hedgers, speculators, arbitrageurs, and market makers.
- Central clearing, margin accounts, daily settlement, default waterfalls, and counterparty risk.
- Contract specification: underlying, size, delivery, maturity, settlement, tick size, and price quotation.
- The no-arbitrage principle as the organizing logic of derivatives pricing.

Mathematical details:

- Payoff notation for a forward: long payoff $S_T - K$, short payoff $K - S_T$.
- Call and put payoffs: $(S_T - K)^+$ and $(K - S_T)^+$.
- Present value and compounding conventions.
- Simple arbitrage inequalities and replication arguments.

In-class applications:

- Build payoff diagrams for forwards, calls, puts, and simple combinations.
- Margin account example for a futures position.
- Discussion: why derivatives can reduce risk for one party while increasing leverage in the system.

Class 2: Hedging with Futures and Determination of Forward/Futures Prices

Hull readings: Chapters 3 and 5; selected examples from Chapter 35 for commodities.

Topics:

- Short hedges and long hedges.
- Basis risk and convergence of futures prices to spot prices.
- Cross hedging and the minimum-variance hedge ratio.
- Stock index futures and beta hedging.
- Stack-and-roll hedging.
- Forward price for investment assets, assets with known income, and assets with known dividend yield.
- Commodity forwards and futures: storage costs, convenience yield, investment assets versus consumption assets.

Mathematical details:

- Hedge ratio $h^* = \rho\sigma_S/\sigma_F$ and optimal number of contracts $N^* = h^* Q_A/Q_F$.
- No-income forward price: $F_0 = S_0 e^{rT}$.
- Known income: $F_0 = (S_0 - I)e^{rT}$.
- Known yield: $F_0 = S_0 e^{(r-q)T}$.
- Forward contract value after initiation: $f = (F_0 - K)e^{-rT}$ for a long forward under constant rates.
- Cost-of-carry form $F_0 = S_0 e^{cT}$ and commodity extension with storage cost and convenience yield.

In-class applications:

- Hedge an equity portfolio with index futures.
- Price a commodity forward with storage cost and convenience yield.
- Explain why futures and expected future spot prices need not be equal.

Class 3: Interest Rates, Bonds, Swaps, and Fixed-Income Derivative Inputs

Hull readings: Chapters 4, 6, and 7.

Topics:

- Simple, annual, and continuous compounding.
- Zero rates, par yields, forward rates, and discount factors.
- Bond pricing and bootstrapping a zero curve.
- Duration and convexity as first- and second-order interest rate risk measures.

- Treasury bond futures, conversion factors, and cheapest-to-deliver intuition.
- SOFR futures and forward rate agreements.
- Plain vanilla interest rate swaps and currency swaps.
- Swap valuation as the difference between fixed-rate and floating-rate bonds.

Mathematical details:

- Discount factor $P(0,T) = e^{-R(0,T)T}$ under continuous compounding.
- Forward rate relation $F_{t_1,t_2} = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}$ for continuously compounded zero rates.
- Bond price $B = \sum_i C_i e^{-r_i t_i}$.
- Duration approximation $\Delta B/B \approx -D\Delta y$ and convexity correction.
- Swap value $V_{\text{swap}} = B_{\text{float}} - B_{\text{fixed}}$ or $B_{\text{fixed}} - B_{\text{float}}$, depending on position.

In-class applications:

- Bootstrap a simple zero curve from bond or swap quotes.
- Value a vanilla fixed-for-floating interest rate swap.
- Discuss why interest rate conventions are operationally important in derivatives markets.

Class 4: Options Markets, Put-Call Parity, Trading Strategies, and Binomial Trees

Hull readings: Chapters 10–13.

Topics:

- Mechanics of options markets: contract terms, moneyness, exercise style, clearing, and margins.
- Determinants of option prices: stock price, strike, time to maturity, volatility, risk-free rate, and dividends.
- Upper and lower bounds for European and American options.
- Put-call parity for non-dividend-paying and dividend-paying stocks.
- Option strategies: covered calls, protective puts, spreads, straddles, strangles, collars, and synthetic forwards.
- One-step and multi-step binomial trees.
- Risk-neutral probabilities and backward induction.
- American option early exercise logic.

Mathematical details:

- Put-call parity for European options: $c + Ke^{-rT} = p + S_0$.
- Dividend-yield version: $c + Ke^{-rT} = p + S_0e^{-qT}$.

- One-step binomial hedge ratio $\Delta = (f_u - f_d)/(S_0 u - S_0 d)$.
- Risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$.
- Backward induction $f = e^{-r\Delta t}[pf_u + (1-p)f_d]$.
- Cox–Ross–Rubinstein parameters $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$.

In-class applications:

- Detect arbitrage violations in option quotes using put-call parity.
- Price a European call and an American put using a binomial tree.
- Compare static option strategies by payoff, cost, and risk exposure.

Week 2: Stochastic Calculus, Black–Scholes–Merton, Greeks, Volatility, and Alternatives

Class 5: Wiener Processes, Itô's Lemma, and Stock Price Dynamics

Hull readings: Chapter 14; review probability and lognormal distribution material from Chapter 15.

Topics:

- Markov processes and why memoryless dynamics are useful in pricing.
- Brownian motion / Wiener process: independent increments, normal increments, and continuous paths.
- Generalized Wiener processes and Itô processes.
- Geometric Brownian motion as the standard stock price model.
- Drift and volatility.
- Correlated stochastic processes.
- Itô's lemma and its financial interpretation.
- Lognormal stock price distribution.

Mathematical details:

- Wiener increment $dz = \epsilon\sqrt{dt}$, with $\epsilon \sim N(0,1)$.
- Itô process $dx = a(x,t)dt + b(x,t)dz$.
- Quadratic variation rule $(dz)^2 = dt$ and higher-order rules $dt\,dz = 0$, $(dt)^2 = 0$.
- Itô's lemma:
$$dG = \left(\frac{\partial G}{\partial x}a + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial x^2}b^2\right)dt + \frac{\partial G}{\partial x}b\,dz.$$
- Geometric Brownian motion $dS = \mu S dt + \sigma S dz$.

- Log price dynamics $d\ln S = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dz$.

In-class applications:

- Apply Itô's lemma to $G = \ln S$ and $G = S^n$.
- Explain why volatility affects expected log returns.
- Simulate one-period and multi-period stock price paths conceptually.

Class 6: Black–Scholes–Merton Pricing and Risk-Neutral Valuation

Hull readings: Chapter 15; optional Chapter 17 for index and currency options; Chapter 18 for Black's model if time permits.

Topics:

- Assumptions of the Black–Scholes–Merton model.
- Delta hedging and construction of a locally riskless portfolio.
- Derivation of the Black–Scholes–Merton partial differential equation.
- Boundary conditions for calls and puts.
- Risk-neutral valuation and change from real-world expected return to risk-free drift.
- Closed-form prices for European calls and puts.
- Dividend yield, index options, currency options, and futures options.
- Implied volatility as the market's translation of prices into model volatility.

Mathematical details:

- Black–Scholes–Merton PDE:

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf.$$

- European call:

$$c = S_0 N(d_1) - Ke^{-rT}N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

- European put:

$$p = Ke^{-rT}N(-d_2) - S_0 N(-d_1).$$

- Dividend yield adjustment: replace S_0 with $S_0 e^{-qT}$ and r with $r - q$ in the stock drift under the risk-neutral measure.
- Risk-neutral expectation:

$$f_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\text{payoff at } T\right].$$

In-class applications:

- Derive the PDE step by step from Itô's lemma and a delta-hedged portfolio.
- Price a European call and put and verify put-call parity.
- Interpret what changes when dividends, foreign interest rates, or futures prices enter the model.

Class 7: Greeks, Dynamic Hedging, Volatility, and Numerical Methods

Hull readings: Chapters 19–21; selected material from Chapters 22–23 if time permits.

Topics:

- Delta, Gamma, Theta, Vega, and Rho.
- Delta hedging in continuous time versus discrete time.
- Gamma and Vega risk; why delta-neutral portfolios remain risky.
- Relationship between Delta, Gamma, and Theta in Black–Scholes–Merton.
- Scenario analysis and stress testing.
- Implied volatility, volatility smiles, volatility skews, and volatility surfaces.
- Estimating volatility from historical data; EWMA and GARCH intuition.
- Numerical pricing overview: binomial trees, Monte Carlo simulation, and finite differences.
- VaR and expected shortfall as risk measures for derivative portfolios.

Mathematical details:

- Delta $\Delta = \partial f / \partial S$.
- Gamma $\Gamma = \partial^2 f / \partial S^2$.
- Theta $\Theta = \partial f / \partial t$.
- Vega $\nu = \partial f / \partial \sigma$.
- Rho $\rho = \partial f / \partial r$.
- Taylor approximation:

$$\Delta f \approx \Delta\, \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \Theta\, \Delta t + \nu\, \Delta\sigma + \rho\, \Delta r.$$

- Black–Scholes–Merton Greek formulas for European calls and puts.
- Historical volatility estimator and exponentially weighted moving average intuition.

In-class applications:

- Construct a delta-neutral portfolio and show why Gamma matters.
- Use Greeks to approximate the effect of a market move.
- Interpret an implied volatility smile as evidence against constant-volatility lognormal pricing.

Class 8: Alternative Investments, Private Equity, Real Options, and Course Integration

Hull readings: Chapter 36; selected material from Chapters 8, 22, 24, 35, and 37. Instructor-supplied private equity notes are required.

Topics:

- What counts as alternative investment: private equity, venture capital, hedge funds, real assets, commodities, infrastructure, real estate, private credit, and structured products.
- Private equity fund structure: limited partners, general partners, management fees, carried interest, hurdle rates, catch-up clauses, fund life, capital calls, and distributions.
- Private equity performance measurement: IRR, money multiple, DPI, RVPI, TVPI, public market equivalent, and the pitfalls of stale valuations.
- Leverage and value creation in buyouts: operating improvement, multiple expansion, tax shields, and debt risk.
- Venture capital and staged financing as compound options.
- Real options: option to delay, expand, abandon, switch inputs, or stage investment.
- Commodity derivatives and real asset hedging.
- Derivatives mishaps: model risk, liquidity risk, basis risk, leverage, incentives, and governance.
- Course synthesis: no-arbitrage, risk-neutral valuation, hedging, volatility, and illiquidity.

Mathematical details:

- IRR as the discount rate solving $\sum_{t=0}^{T} CF_t/(1+\mathrm{IRR})^t = 0$.
- Money multiple = total distributions and residual value/paid-in capital.
- Simple carried interest waterfall calculations.
- Real option valuation by binomial tree and risk-neutral probabilities.
- Mapping investment flexibility to option payoffs: delay as a call option, abandonment as a put-like floor, expansion as a growth option.
- Illiquidity and stale-pricing bias in measured volatility and correlation.

In-class applications:

- Evaluate a staged venture investment using a binomial real-options framework.
- Compute basic private equity fund performance measures from cash flows.
- Discuss how derivative risk-management logic applies, and where it fails, in illiquid alternative investments.

Suggested Problem Practice

Students should attempt end-of-chapter practice questions from Hull after each class. The instructor may assign a narrower list depending on the pace of the course.

After Class 1	Hull Chapters 1–2: market structure, payoff diagrams, margin, and contract mechanics.
After Class 2	Hull Chapters 3 and 5: hedge ratios, basis risk, and forward/futures pricing.
After Class 3	Hull Chapters 4, 6, and 7: zero rates, forward rates, bond pricing, duration, and swaps.
After Class 4	Hull Chapters 10–13: put-call parity, option bounds, option strategies, and binomial trees.
After Class 5	Hull Chapter 14: Brownian motion, Itô processes, Itô's lemma, and lognormality.
After Class 6	Hull Chapter 15: BSM PDE, risk-neutral valuation, closed-form formulas, and implied volatility.
After Class 7	Hull Chapters 19–21 and 23: Greeks, hedging, volatility surfaces, numerical methods, and volatility estimation.
After Class 8	Hull Chapters 35–37 plus instructor notes: real options, commodities, private equity, and derivative failures.

Mathematical Emphasis

The course will treat the following derivations as central:

1. No-arbitrage derivation of forward prices.
2. Minimum-variance hedge ratio for futures hedging.
3. Put-call parity and option price bounds.
4. Binomial option pricing by replication and by risk-neutral valuation.
5. Itô's lemma for functions of stochastic processes.
6. Lognormal stock price dynamics under geometric Brownian motion.
7. Black–Scholes–Merton PDE from a delta-hedged portfolio.
8. Black–Scholes–Merton formula by risk-neutral valuation.
9. Greek-based approximation of changes in option values.
10. Real-options interpretation of staged investment and abandonment decisions.

Closed-Book Exam Expectations

Students should be prepared to:

- Define key instruments and explain their economic use.

- Draw and interpret payoff diagrams.
- Solve numerical pricing and hedging problems without access to notes.
- Derive core no-arbitrage and Black–Scholes–Merton equations.
- Apply Itô's lemma to simple functions.
- Compute option prices and Greeks when formulas are provided or required from memory.
- Interpret results economically, including the limitations of the assumptions used.
- Explain how alternative investments differ from liquid traded derivatives in valuation, risk measurement, and governance.

Course Philosophy

Derivatives are best learned by moving repeatedly between three views: contract mechanics, no-arbitrage mathematics, and economic interpretation. Students should therefore expect each class to combine institutional details, derivations, numerical examples, and discussion of real market use. The mathematical parts of the course are essential: they explain why derivative prices are linked across instruments, why hedging can create valuation equations, and why model assumptions matter in practice.